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Paper 1: Optimizing Ibrutinib Posology in Chronic Lymphocytic Leukemia Using a Semi-Mecha

Bibliographic Information

- **Title:** Optimizing Ibrutinib Posology in Chronic Lymphocytic Leukemia Using a Semi-Mechanistic Pharmacometric Framework
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- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2025
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Quick Take

This paper presents a comprehensive semi-mechanistic PKPD framework for ibrutinib in CLL, integrating efficacy (SPD, spleen volume, leukocyte/lymphocyte counts) and safety (blood pressure) biomarkers. The model quantifies key differences between treatment-naïve and relapsed/refractory patients and demonstrates through simulations that dose reductions guided by response depth or toxicity can maintain comparable efficacy while reducing hypertension risk. This work is essential reading for pharmacometricians and clinical pharmacologists involved in dose optimization for oncology drugs, particularly those interested in model-informed drug development and personalized medicine approaches.

Executive Summary

This study extends a previously developed semi-mechanistic PKPD framework for ibrutinib in CLL by integrating Phase 3 data from treatment-naïve patients, adding spleen volume and lymphocyte count as efficacy biomarkers, and quantifying differences between TN and R/R populations. The model simultaneously characterizes the time course of phosphorylated Btk, CLL cell dynamics in lymphoid tissues and peripheral blood, and blood pressure changes. Key findings include a 76% longer pBtk half-life and 43% shorter peripheral CLL cell half-life in TN versus R/R patients, with no evidence of resistance development in TN patients. Simulation-based evaluation of alternative dosing strategies demonstrated that dose reductions guided by response depth ($\geq 50\%$ reduction in SPD and lymphocyte count) or toxicity (Grade 3 hypertension) maintain comparable response rates and PFS to standard 420 mg/day dosing, while reducing hypertension risk. In combination with venetoclax, ibrutinib de-escalation schedules resulted in $\leq 5\%$ reduction in peripheral blood MRD-negative rates at 2 years. This framework provides a quantitative foundation for dose individualization in CLL, supporting improved tolerability without compromising efficacy.

Scientific Context & Motivation

Ibrutinib, a first-in-class BTK inhibitor, is a cornerstone therapy for CLL but is associated with significant toxicities (hypertension, atrial fibrillation, arthralgia) that lead to dose reductions or discontinuation in 20-50% of patients in real-world practice. Emerging evidence suggests that reduced doses (280 or 140 mg/day) may retain clinical efficacy while improving tolerability. However, the optimal dosing strategy, particularly in combination with venetoclax, remains poorly defined. Previous modeling work by the same group established a semi-mechanistic framework in R/R patients, but lacked data from TN patients and did not include spleen volume or lymphocyte count as endpoints. This study addresses these gaps by integrating Phase 3 data from TN patients, extending the biomarker set, and quantifying disease-state differences. The work aligns with the FDA's recent guidance on dose optimization in oncology, which emphasizes the need for exposure-response analyses to identify doses that maximize efficacy while minimizing toxicity.

✂ Methodological Snapshot

The study employed a semi-mechanistic population PKPD modeling framework that integrates multiple efficacy and safety biomarkers. The model comprises a pBtk compartment (target engagement), four CLL cell subpopulations (two proliferating in lymphoid tissues, one detaching, one in peripheral blood), and a blood pressure turnover model with transit compartment. Ibrutinib's effects are modeled as inhibition of pBtk production (Imax model) and stimulation of cell detachment from stroma (linear slope). Disease state (TN vs R/R) was identified as a key covariate on multiple parameters. The model was developed using nlmixr with FOCEI estimation and evaluated via VPCs. Simulations of 1000 virtual patients were used to evaluate response-based dose modification, toxicity-adjusted dosing, and combination with venetoclax.

☒ Structural Model Breakdown

The model comprises several interconnected components: (1) A population PK model (from Marostica et al.) that computes daily AUC of ibrutinib. (2) A pBtk compartment representing the phosphorylated form of the target, with turnover described by a zero-order production rate (k_{in_pBtk}) and first-order elimination (k_{out_pBtk}). Ibrutinib inhibits pBtk production via an Imax model: $Effect = 1 - (Imax * AUC) / (AUC_{50} + AUC)$. (3) Four CLL cell subpopulations: two proliferating colonies attached to stroma in lymphoid tissues (Prolif1, Prolif2), a third subtype that detaches from stroma and leaves lymphoid tissues (Detach), and a fourth subtype in peripheral blood (Blood). The proliferation rate (k_{prolif}) is shared across the two tissue colonies. Detachment of cells from stroma is stimulated by ibrutinib via a linear slope model: $k_{det} = k_{det0} * (1 + slope * AUC)$. (4) Cells in peripheral blood undergo natural death (k_{death}) and can home back to lymphoid tissues (k_{home}). (5) For combination with venetoclax, an additional killing effect is applied to CLL cells in both tissues and blood, with higher potency in blood (Emax model with different EC50 values). (6) The blood pressure model uses a turnover model with a single transit compartment. Ibrutinib stimulates the zero-order production rate of sBP and dBP via an Emax

model: $\text{Stim} = 1 + (\text{Emax} * \text{AUC}) / (\text{AUC50_BP} + \text{AUC})$. The model includes a shared Emax for sBP and dBP, with separate baseline values and AUC50 parameters.

Detailed Methodological Analysis

Modeling Approach

A semi-mechanistic population PKPD modeling framework was developed, extending a previously published model. The model includes: (1) A population PK model for ibrutinib (from Marostica et al.) to compute daily AUC. (2) A pBtk compartment representing the target engagement. (3) Four CLL cell subpopulations: two proliferating colonies attached to stroma in lymphoid tissues, a third subtype that detaches and leaves tissues, and a fourth subtype in peripheral blood. (4) Ibrutinib inhibits pBtk production (Imax model) and stimulates detachment of cells from stroma (linear slope model). (5) A turnover model with transit compartment for blood pressure, where ibrutinib stimulates the zero-order production rate of sBP and dBP (Emax model). (6) For combination with venetoclax, an additional killing effect was applied to CLL cells in both tissues and blood, with higher potency in blood (Emax model).

Data Sources

Data were obtained from two clinical trials: a phase 1b-2 study (PCYC-1102, n=132) and a phase 3 study (PCYC-1115, n=136). The analysis included patients with available ibrutinib plasma concentrations, leukocyte/lymphocyte counts, SPD or spleen volume measurements, and blood pressure data. Ibrutinib was administered orally at 420 mg/day (n=94) or 840 mg/day (n=38) in the phase 1b-2 study, and at 420 mg/day in the phase 3 study. The dataset was accessed through the Yale University Open Data Access Project.

Estimation Methods

Nonlinear mixed-effects modeling was performed using the nlmixr R package (version 2.0.6) with the first-order conditional estimation method with interaction (FOCEI). Statistical significance was set at $p < 0.001$.

Model Evaluation

Model evaluation was performed using visual predictive checks (VPCs) based on 400 simulated datasets. Parameter uncertainty was assessed via 95% confidence intervals and coefficient of variation (CV%). Shrinkage was reported for random effects. The model was evaluated for its ability to simultaneously capture the temporal dynamics of all biomarkers (SPD, spleen volume, leukocyte count, lymphocyte count, sBP, dBP).

Covariate Analysis

Disease state (TN vs R/R) was identified as a significant covariate on several parameters: pBtk turnover rate (1.76-fold higher in R/R), baseline peripheral CLL cell count (4-fold higher in TN), normal leukocyte count (47% lower in R/R), baseline SPD (58% higher in R/R), and peripheral CLL cell death rate (43% shorter half-life in TN). Age was included as a covariate on baseline diastolic blood pressure. Covariate selection was based on statistical significance ($p < 0.001$) and

clinical relevance. No other covariates (e.g., sex, renal function, concomitant medications) were evaluated, which is a limitation.

Statistical Rigor Assessment

The modeling approach is methodologically sound, using nonlinear mixed-effects modeling with FOCEI estimation. The sample size (n=246 with PK and efficacy data) is adequate for the model complexity, though the number of patients with blood pressure data is not explicitly stated. Parameter uncertainty is reported with 95% CIs and CV%, and most parameters are estimated with acceptable precision. However, some random effects show high shrinkage (e.g., 53% for k_{det2} , 44% for k_{out_pBtk}), indicating limited ability to estimate individual-level parameters for these processes. The statistical significance threshold of $p < 0.001$ is appropriately conservative for covariate selection. VPCs based on 400 simulations provide adequate evaluation of model performance. The study does not report formal sensitivity analyses or bootstrap evaluations, which would strengthen the assessment of parameter identifiability and model robustness. Missing data handling is not explicitly discussed, though the analysis includes only patients with available data. The simulation scenarios are well-designed with 1000 virtual patients, providing adequate precision for comparing dosing strategies.

Key Findings

The study quantified several key differences between TN and R/R CLL patients: (1) TN patients had a 76% longer pBtk half-life (k_{out_pBtk} : 0.310 vs 0.524 day^{-1} for R/R), indicating slower turnover of the target. (2) TN patients had a 43% shorter peripheral CLL cell half-life (k_{death} : 0.162 vs 0.124 day^{-1} for R/R), suggesting faster elimination of circulating cells. (3) No evidence of ibrutinib resistance development was observed in TN patients within the analyzed timeframe. (4) Baseline peripheral blood CLL cell counts were approximately 4-fold higher in TN patients (3279 vs 208 cells/ μL for R/R). (5) SPD at baseline was 58% higher in R/R patients (314 vs 252 cm^2 for TN), consistent with higher incidence of bulky disease. (6) For response-based dose modification, dose reduction to 140 mg/day after achieving $\geq 50\%$ response depth maintained comparable response rates (excessive relative risk of non-response $< 3\%$) and PFS (relative risk 0.9-1.0) to standard dosing, while reducing hypertension risk by $\sim 18\%$. (7) Toxicity-adjusted dosing based on Grade 3 hypertension provided a favorable risk-benefit ratio (0.47 for sBP, 0.24 for dBP) with minimal impact on PFS. (8) In combination with venetoclax, ibrutinib de-escalation (420 $\rightarrow$ 280 $\rightarrow$ 140 mg/day) resulted in MRD-negative rates of 58% (TN) and 31% (R/R) at 2 years, compared to 63% and 34% with standard dosing.

Clinical & Regulatory Implications

The model supports several clinically actionable strategies: (1) Dose reduction to 140 mg/day after achieving a $\geq 50\%$ reduction in lymph node size and lymphocyte count maintains comparable response rates and PFS to standard dosing, while reducing hypertension risk by $\sim 18\%$. (2) Toxicity-adjusted dosing based on Grade 3 hypertension (sBP ≥ 160 or dBP ≥ 100 mmHg) provides a

favorable risk-benefit balance, with modest reductions in hypertension risk and minimal impact on response. (3) In combination with venetoclax, stepwise ibrutinib de-escalation (420→280→140 mg/day) results in $\leq 5\%$ reduction in MRD-negative rates at 2 years, supporting time-limited combination therapy. (4) The framework quantifies key differences between TN and R/R patients: TN patients have 76% longer pBtk half-life and 43% shorter peripheral CLL cell half-life, contributing to better outcomes. These findings have direct regulatory implications for dose optimization in oncology, aligning with FDA guidance on maximizing efficacy while enhancing tolerability.

Strengths & Limitations

Strengths

- Comprehensive integration of multiple efficacy and safety biomarkers within a single mechanistic framework.
- Quantification of key differences between TN and R/R patients, enabling population-specific dosing recommendations.
- Clinically relevant simulation scenarios covering response-based, toxicity-based, and combination therapy approaches.
- Use of a semi-mechanistic model structure that is biologically plausible and supports extrapolation beyond observed data.
- Simultaneous modeling of sBP and dBP with shared parameters, improving statistical efficiency.
- Alignment with FDA guidance on dose optimization in oncology, providing a regulatory-relevant framework.
- Open data access through YODA project enhances transparency and reproducibility.

Limitations (Acknowledged by Authors)

- The modeling framework was developed using clinical trial data, which may lead to underestimating expected toxicities in real-world clinical settings.
- The analyzed data had an average follow-up of 2 years, so extrapolations beyond this period should be interpreted cautiously.
- The model simplified the assessment of resistance evolution in CLL due to the short study duration and absence of molecular resistance assays.
- The evaluation of combination treatment should be considered carefully, as the effect of venetoclax on CLL dynamics was quantified and externally validated using digitized sources.
- MRD values of $\leq 0.001\%$ in peripheral blood are assay-dependent.

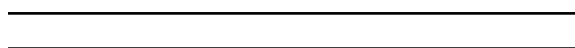
Limitations (Expert Review)

- The model does not include other important toxicities such as atrial fibrillation, arthralgia, or bleeding risk, which are clinically relevant for dose decisions.
- No formal sensitivity analysis or bootstrap evaluation was performed to assess parameter identifiability and model robustness.
- The high shrinkage on several random effects (e.g., 53% for k_{det2}) limits the reliability of individual-level predictions.

- The covariate analysis was limited to disease state and age; other potentially important covariates (e.g., renal function, concomitant medications, genetic factors) were not evaluated.
- The venetoclax effect model lacks inter-individual variability, which may underestimate the variability in combination therapy outcomes.
- The model assumes a fixed relationship between daily AUC and effect, without considering potential time-dependent changes in exposure or tolerance development.
- The response-based dose modification scenarios assume perfect adherence to monitoring schedules, which may not reflect real-world practice.

Generalizability

The model was developed using clinical trial data, which may underestimate toxicities seen in more diverse real-world populations with comorbidities and concomitant medications. The average follow-up of 2 years limits extrapolation to longer-term outcomes. The findings are specific to ibrutinib in CLL and may not generalize to other BTK inhibitors or indications. The combination therapy simulations are based on digitized data for venetoclax and lack inter-individual variability, limiting quantitative precision.



Figures & Tables

- **Figure 1:** Schematic of the semi-mechanistic PKPD framework showing the relationship between ibrutinib PK, pBtk dynamics, CLL cell subpopulations (proliferating, detached, peripheral), and blood pressure turnover model.
 - *Significance:* Provides a visual overview of the integrated model structure, including the four CLL cell subpopulations, the effect of ibrutinib on pBtk production and cell detachment, and the blood pressure model with transit compartment.
- **Figure 2:** Prediction-corrected VPCs for the PK-SPD-spleen-leukocyte-lymphocyte model (A, B) and the PK-blood pressure model (C), showing observed data (gray points) with median and 90% prediction intervals.
 - *Significance:* Demonstrates the model's predictive performance across all biomarkers, indicating adequate fit to the observed data for both efficacy and safety endpoints.
- **Figure 3:** Excessive relative risk of non-response and hypertension (sBP ≥ 140 , dBP ≥ 90 mmHg) at 2 years for response-based dose modification scenarios, comparing treatment cessation and dose reduction to 140 mg/day across different initial dosing periods (3-12 months) and response depth criteria (50%, 75%).
 - *Significance:* Quantifies the trade-off between efficacy and safety for different response-guided dosing strategies, showing that dose reduction maintains efficacy while reducing hypertension risk.
- **Figure 4:** Excessive relative risk of non-response and hypertension for toxicity-adjusted dosing scenarios based on Grade 2/3 hypertension and relative change from baseline sBP/dBP thresholds.
 - *Significance:* Evaluates the risk-benefit of toxicity-guided dose adjustments, showing that Grade 3 hypertension-adjusted dosing provides a favorable balance with modest efficacy loss.

- **Figure 5:** Proportions of MRD responses in peripheral blood at 2 years for ibrutinib monotherapy, venetoclax monotherapy, and combination therapy with standard or de-escalated ibrutinib dosing.
 - *Significance:* Demonstrates that ibrutinib de-escalation (420→280→140 mg/day) in combination with venetoclax results in only $\leq 5\%$ reduction in MRD-negative rates compared to standard dosing, supporting time-limited combination therapy.
 - **Table 1:** Parameter estimates for the PK-SPD-spleen-leukocyte-lymphocyte model, including fixed effects, random effects (CV%), and residual unexplained variability.
 - *Significance:* Provides the quantitative basis for all simulation scenarios, including key parameters such as pBtk turnover rate, CLL cell proliferation rate, and the effect of disease state (TN vs R/R) on these parameters.
 - **Table 2:** Parameter estimates for the PK-blood pressure model, including baseline sBP and dBP, transit time, maximum stimulatory effect, and AUC50.
 - *Significance:* Quantifies the relationship between ibrutinib exposure and blood pressure increase, enabling simulation of hypertension risk under different dosing strategies.
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Code & Reproducibility Assessment

The dataset was made available through the Yale University Open Data Access Project (YODA Project 2020-4386). The nlmixr R package (version 2.0.6) was used for model development with FOCEI estimation, and RxODE (version 1.1.1) for simulations. However, no publicly accessible code repository (e.g., GitHub) is mentioned, limiting full reproducibility.

Future Directions

Several important questions remain: (1) Prospective validation of the proposed dosing strategies in clinical trials is essential before clinical implementation. (2) The model should be extended to include longer-term follow-up data (>2 years) to assess durability of response and late-emerging toxicities. (3) Incorporation of molecular resistance assays (e.g., BTK C481S mutations) would improve the model's ability to predict resistance development. (4) The venetoclax component should be refined with individual patient data and inter-individual variability to improve combination therapy predictions. (5) Evaluation of venetoclax dose reduction strategies in combination with ibrutinib is warranted to minimize adverse effects and improve cost-effectiveness. (6) The framework could be extended to include other BTK inhibitors (e.g., acalabrutinib, zanubrutinib) to compare their benefit-risk profiles. (7) External validation using real-world data would strengthen the generalizability of the findings. (8) The model could be used to design adaptive clinical trials with response- or toxicity-based dose modification rules.

Expert Commentary

This work represents a mature application of semi-mechanistic PKPD modeling to address a clinically pressing problem: dose optimization of ibrutinib in CLL. The authors have built upon their prior framework (which was primarily in R/R patients) and extended it to TN patients, incorporating additional biomarkers (spleen volume, lymphocyte count) and quantifying disease-

state differences. The modeling approach is sound, using a joint model for multiple efficacy endpoints (SPD, spleen volume, leukocyte, lymphocyte) with shared pBtk dynamics, which is both biologically plausible and parsimonious. The inclusion of a blood pressure turnover model to capture hypertension as a key adverse event is a strength, enabling simultaneous evaluation of efficacy-safety trade-offs. From a methodological perspective, the use of FOCEI in nlmixr is appropriate for this nonlinear mixed-effects model, though the complexity of the model (multiple ODEs, latent variables) raises questions about parameter identifiability and the adequacy of FOCEI versus SAEM or Bayesian approaches. The authors report acceptable parameter uncertainty (95% CIs), but the high shrinkage on some random effects (e.g., 53% for k_{det2}) warrants caution in interpreting individual-level predictions. The simulation scenarios are clinically relevant and well-designed, covering response-based dose modification, toxicity-adjusted dosing, and combination with venetoclax. The finding that dose reductions maintain efficacy while reducing hypertension is consistent with real-world evidence and supports the FDA's dose optimization initiative. However, the model's reliance on clinical trial data may underestimate toxicities seen in more diverse real-world populations. The combination therapy simulations, while promising, are based on digitized data for venetoclax effects and lack inter-individual variability in venetoclax response, limiting their quantitative precision. Overall, this framework is a valuable tool for hypothesis generation and trial design, but prospective validation in clinical studies is essential before clinical implementation.

Bottom Line

This semi-mechanistic PKPD framework, integrating efficacy (SPD, spleen volume, leukocyte/lymphocyte counts) and safety (blood pressure) biomarkers, demonstrates that ibrutinib dose reductions guided by response depth or toxicity can maintain comparable response rates and progression-free survival to standard 420 mg/day dosing in both treatment-naïve and relapsed/refractory CLL patients. The model also shows that de-escalation of ibrutinib in combination with venetoclax has minimal impact on peripheral blood MRD at 2 years. These findings provide a quantitative basis for dose individualization strategies to improve tolerability without sacrificing efficacy, supporting a more personalized approach to CLL management.

☒ Figures

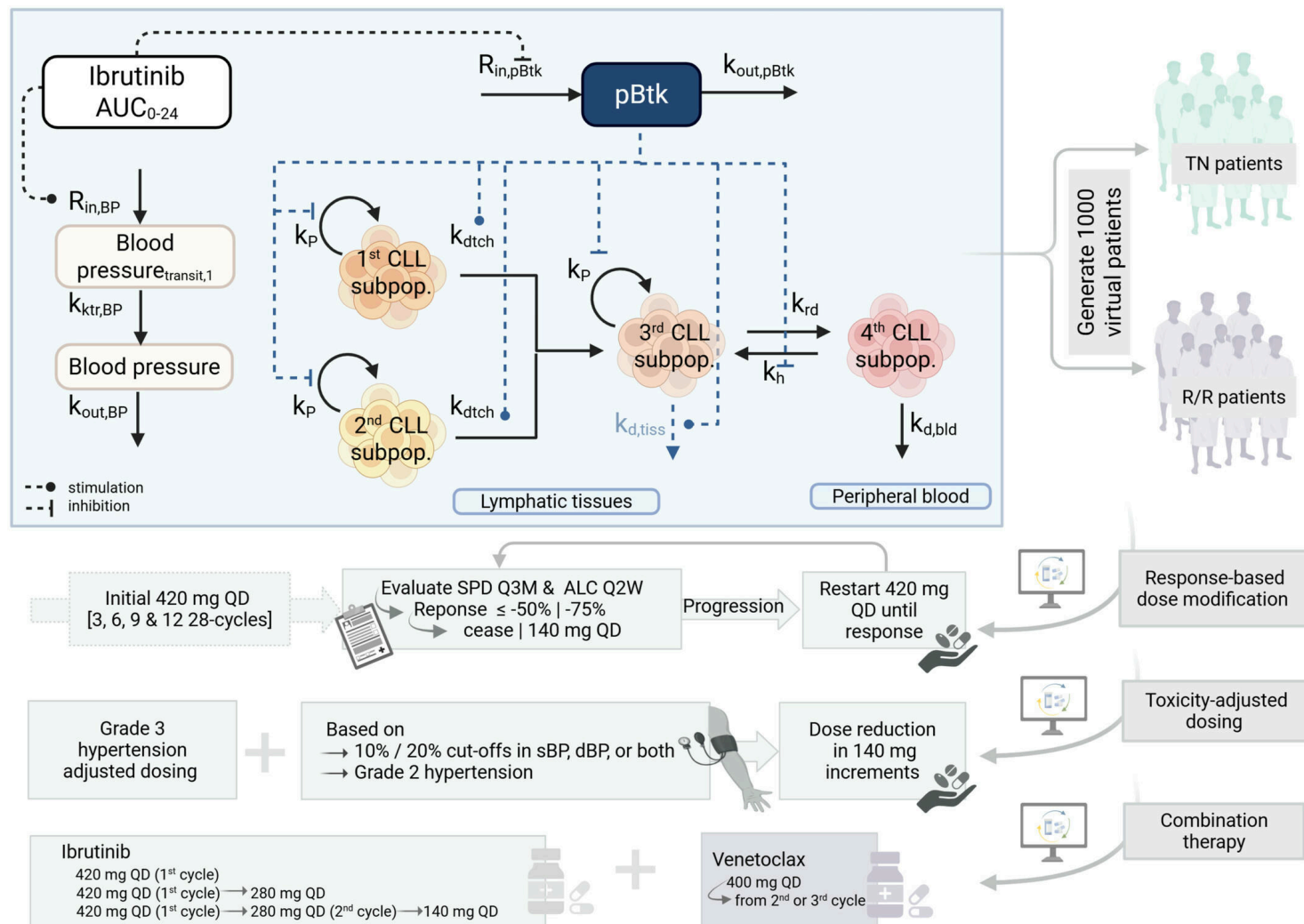


Figure 1: Schematic representation of the population semi-mechanistic PK-PD modeling framework for ibrutinib in chronic lymphocytic leukemia patients, along with the simul

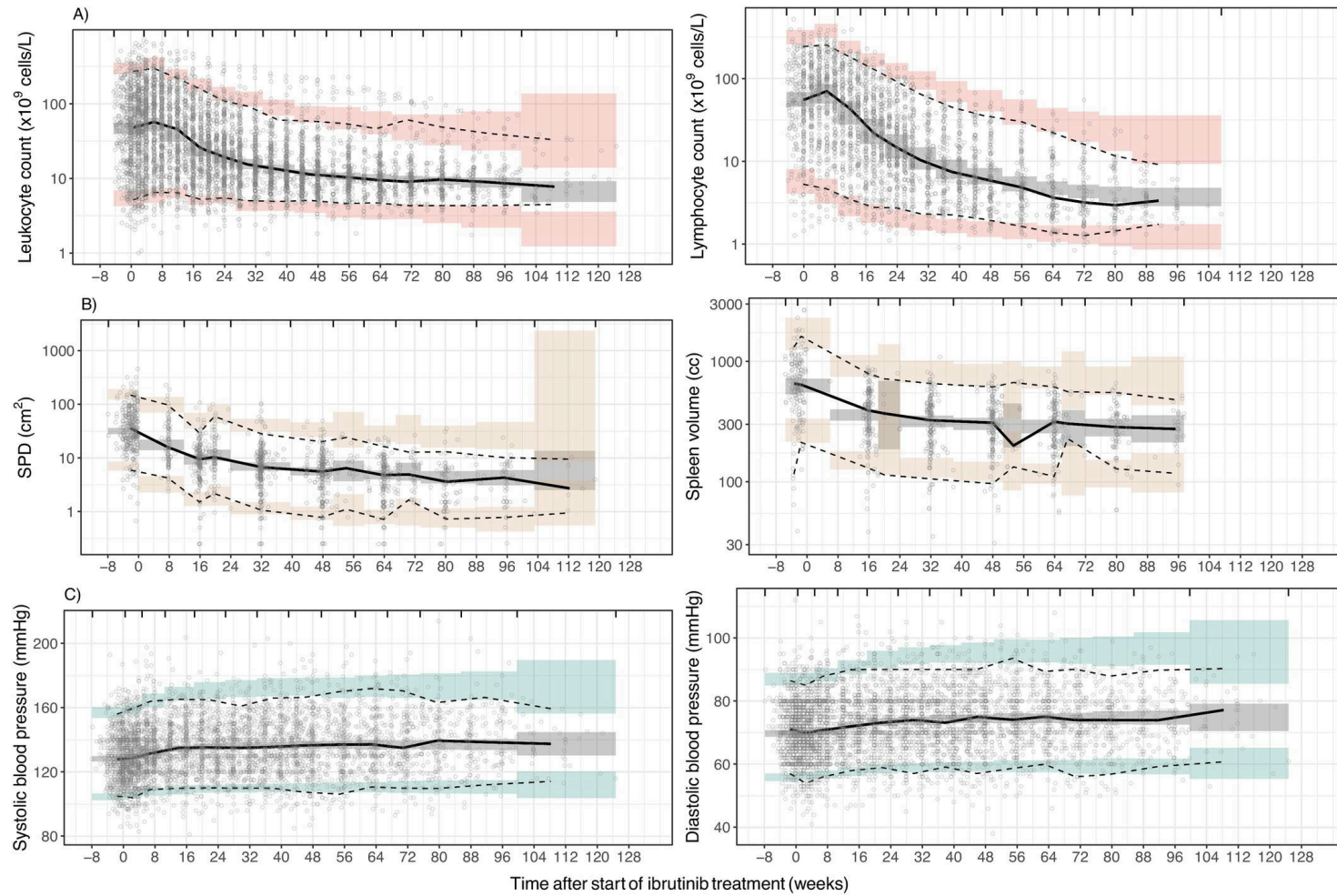


Figure 2: Prediction corrected visual predictive checks of the PK–SPD–spleen–leukocyte–lymphocyte model (A, B) and the PK–blood pressure models (C). Gray points are the ob

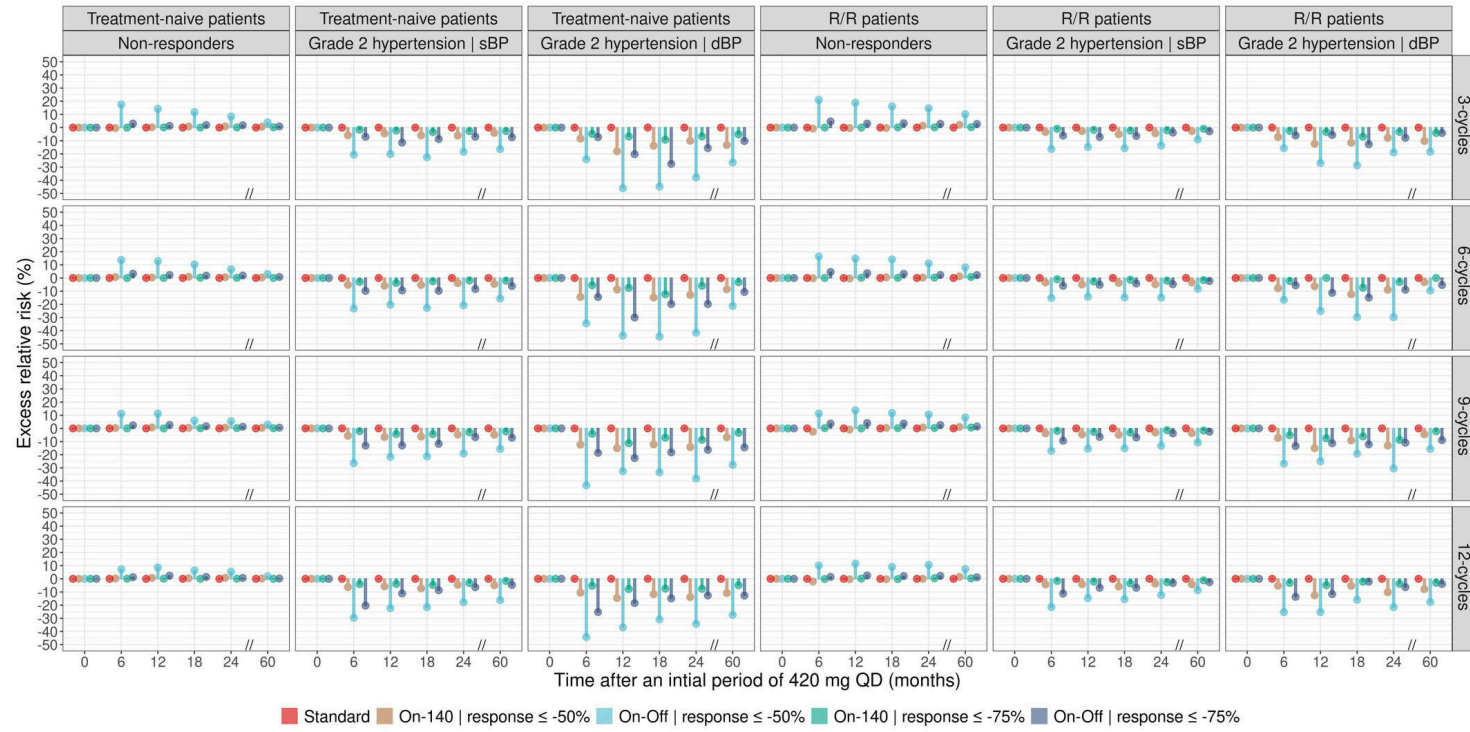


Figure 3: Response-based dose modification simulations from the PK–SPD–spleen–leukocyte–lymphocyte model and the PK-blood pressure models, illustrated as the excessive rel

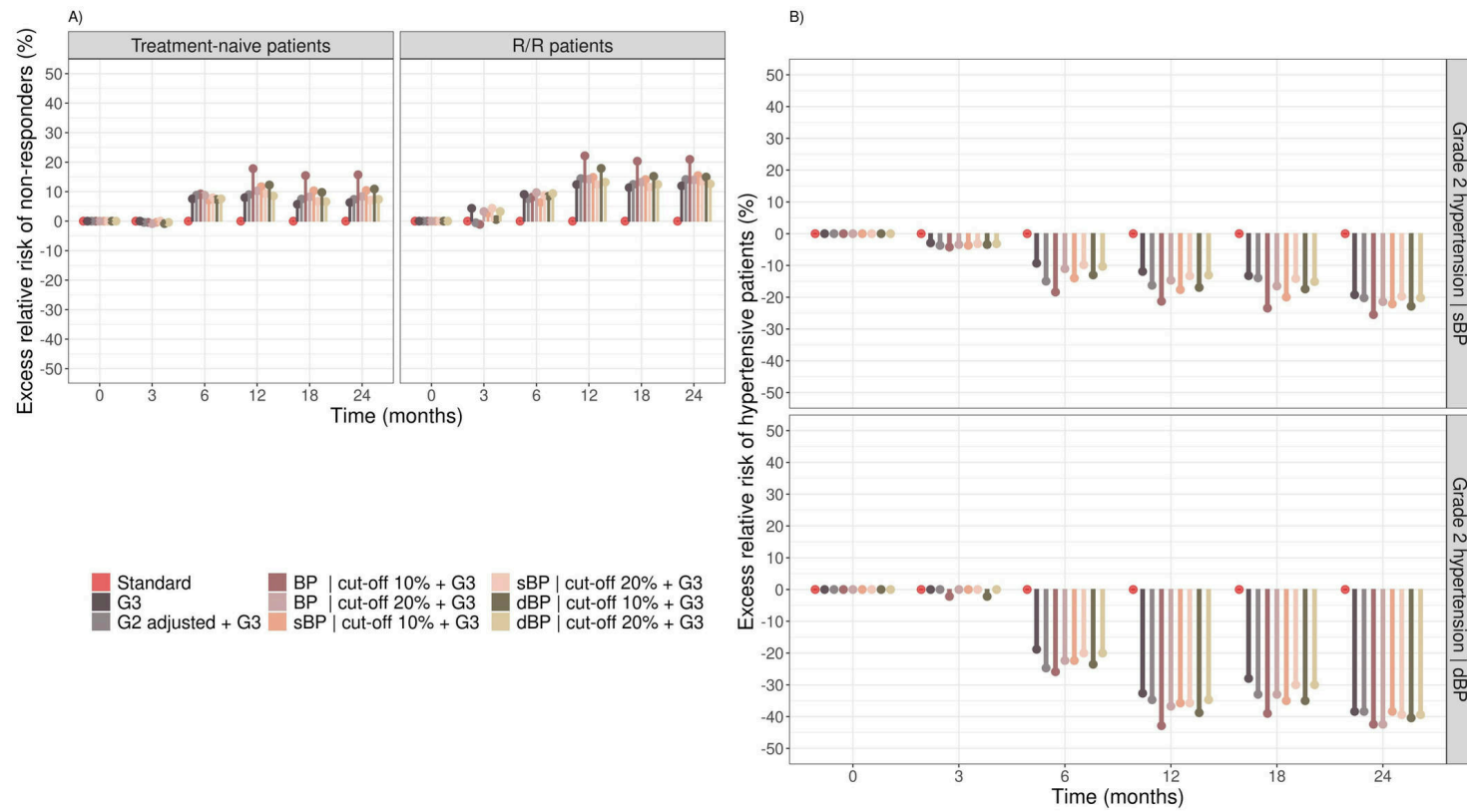


Figure 4: Toxicity-adjusted dosing simulations from the PK–SPD–spleen–leukocyte–lymphocyte model (A) and the PK–blood pressure models (B), illustrated as the excessive rel

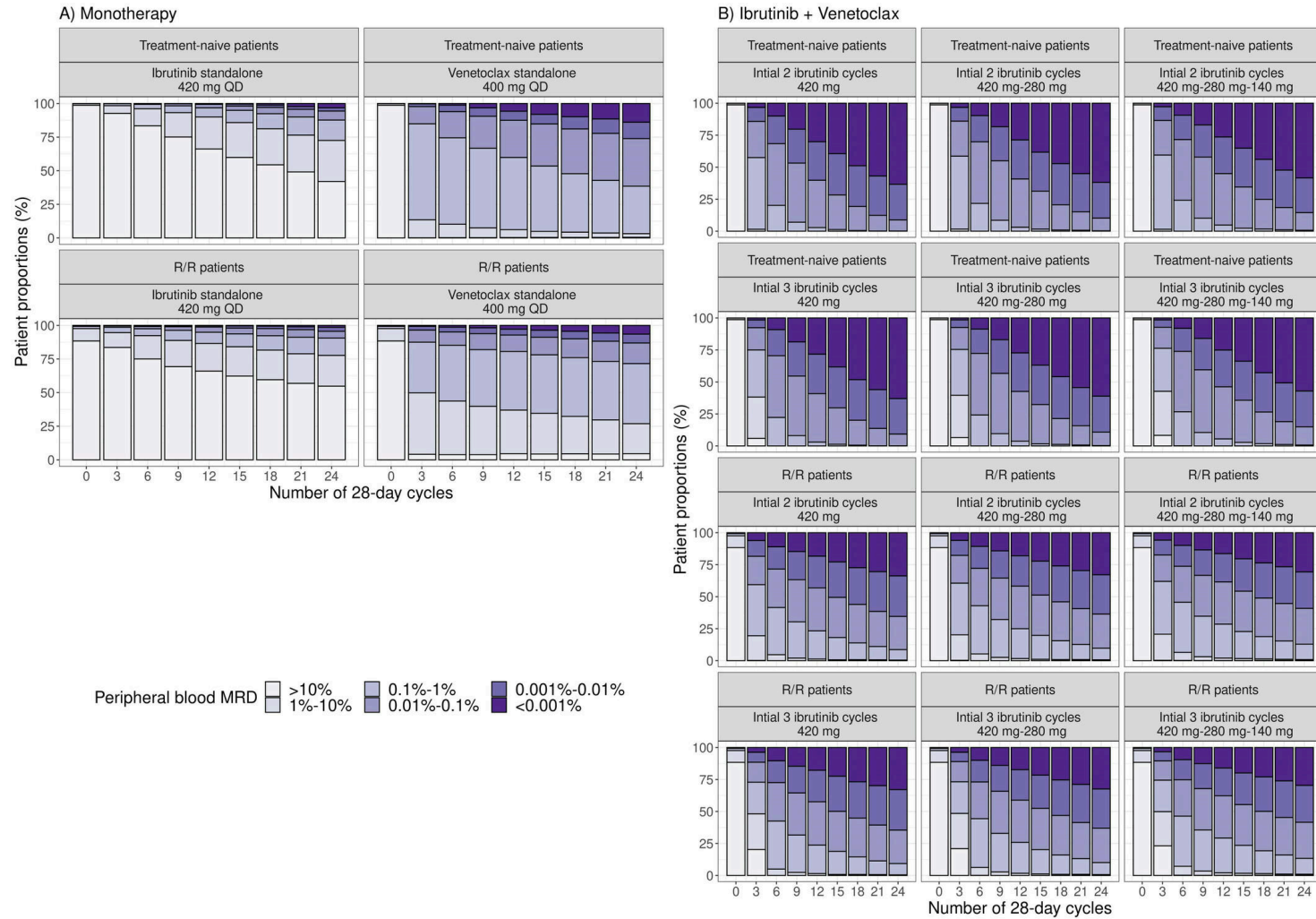


Figure 5: Simulations from the PK-SPD-spleen-leukocyte-lymphocyte model, showing the proportions of the different measurable residual disease responses in peripheral blood